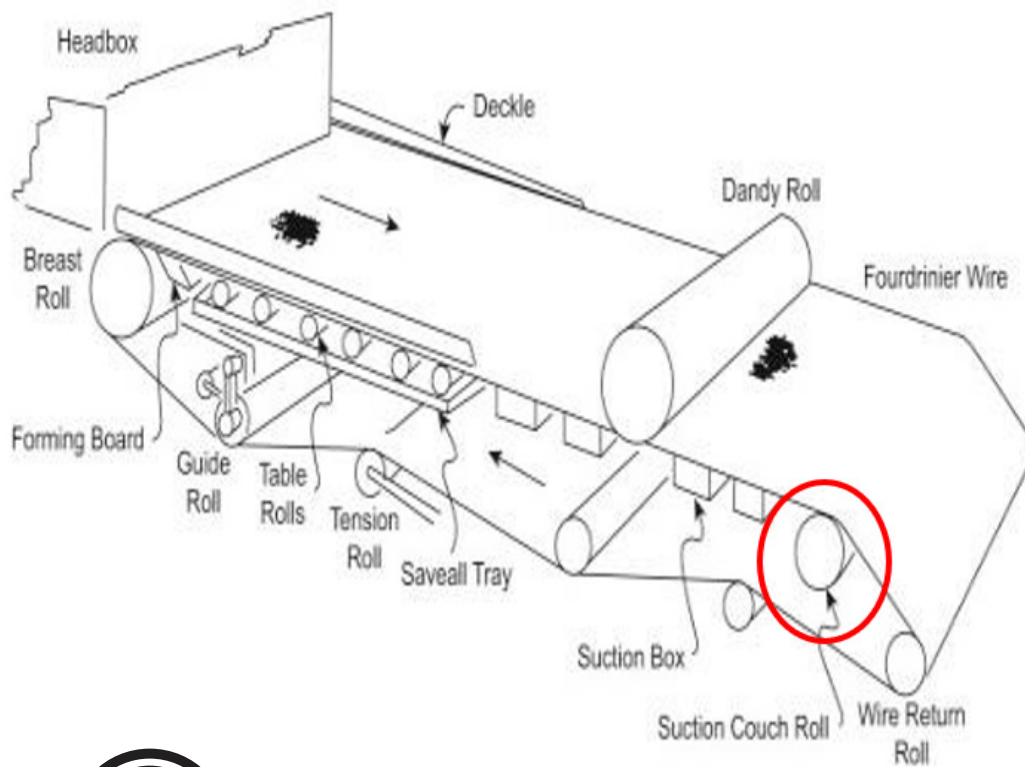


SEALING STRIPS

For Suction Couch Roll

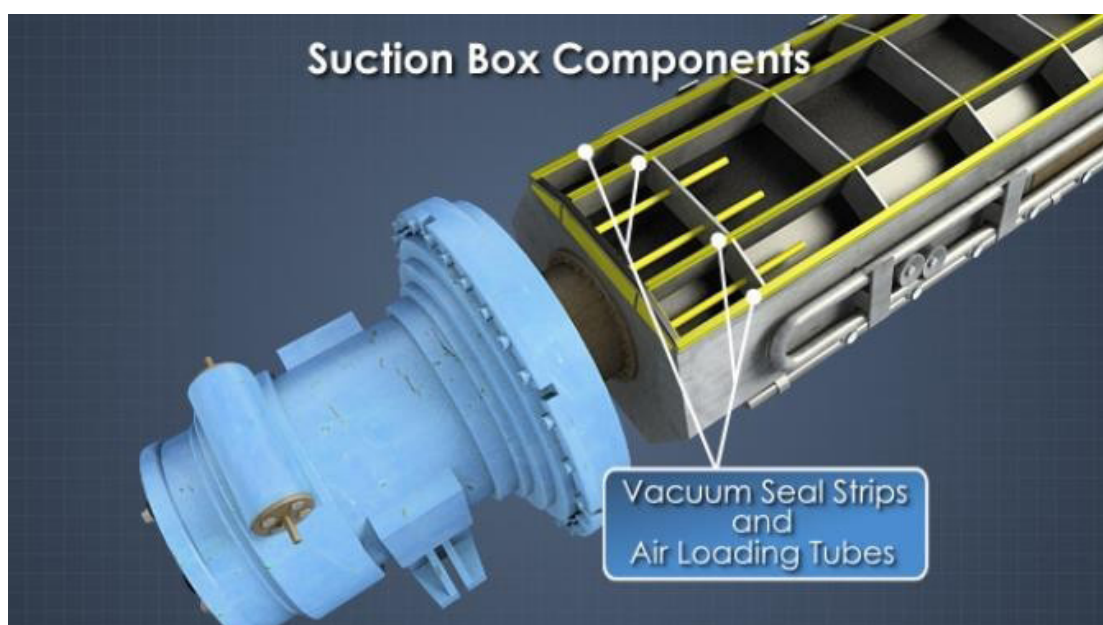
LOCATION & FUNCTION OF SUCTION COUCH ROLL

- As shown in the Fig 1 at the end of the drainage table is the suction couch roll.
- Nowadays 2 nos. of rolls due to high speed, forms the end of the drainage section of a paper making machine
- The function of the suction couch roll is to remove water from the paper by application of vacuum.



SUCTION COUCH ROLL

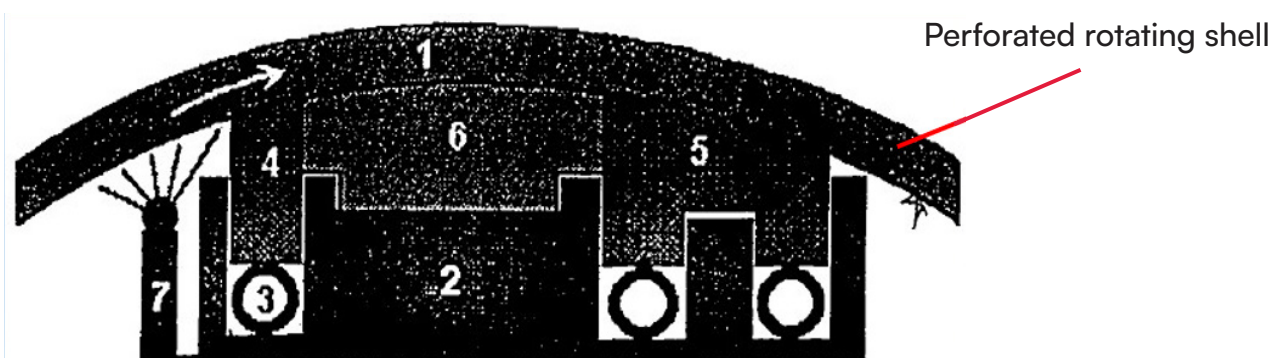
The suction roll consist of 2 primary sections as shown in the Fig 2 below. The shell rotates around the vacuum box and vacuum through the pores pulls water from the paper and its carrying wire or felt as a mating roll press against the diameter of the shell at this point.



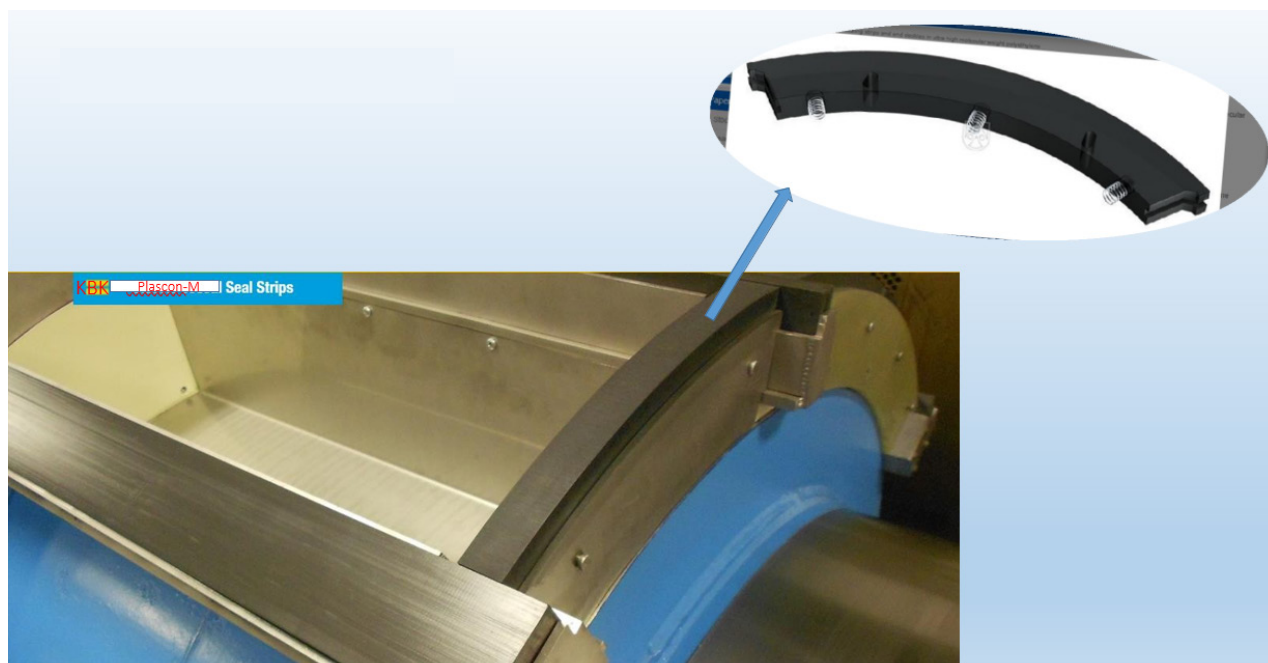
TYPES OF SEALS IN SUCTION COUCH ROLL



Fig 3 shows roll shell (1) with types of packing. Number 4, 5 and 6 are side seal, silencer seal and end deckle respectively, which are made of Plascon-M. Number 7 in the figure below is called lube/fog shower, used to clean the gap between the seals and roll shell. Rest can be read in Fig 5.



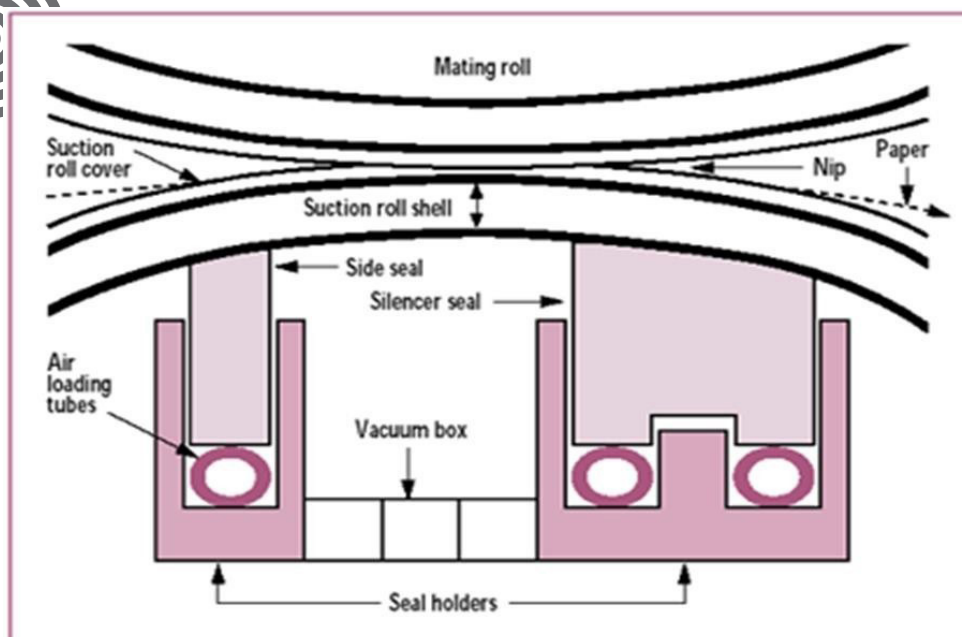
SEALING STRIPS



Greyish black colour (3 nos.) Plascon-M sealing strips
seating to rough roll

WHY ARE SEALING STRIPS IMPORTANT?

- For the vacuum opening to be sealed to transfer the vacuum through the holes in the rotating shells, a seal is required between the stationary box and rotating shell.
- Seals are crucial elements in functioning of suction roll because if it fails the vacuum in the roll is lost.
- The inefficiency of seals can result in improper functioning of suction roll, that is less moisture removal.
- Seal inefficiency will also cause high energy consumption as suction roll will require more vacuum to suck the water out from paper.



Inside the suction couch roll

WHY SEALING STRIPS WEAR OUT

- Finish of interior surface of shell
- Shell or box deflection
- Pressure in air loading tubes (Fig 3, Number 3) should be 4-6 psi
- Thermal expansion
- Dirt trap
- Incorrect machining of shell or seal.
- Pressure of showers (Fig 3, Number 7) should range from 25- max 45 psi.
- Incorrect material selection of sealing strips



DIRT TRAP WEARING

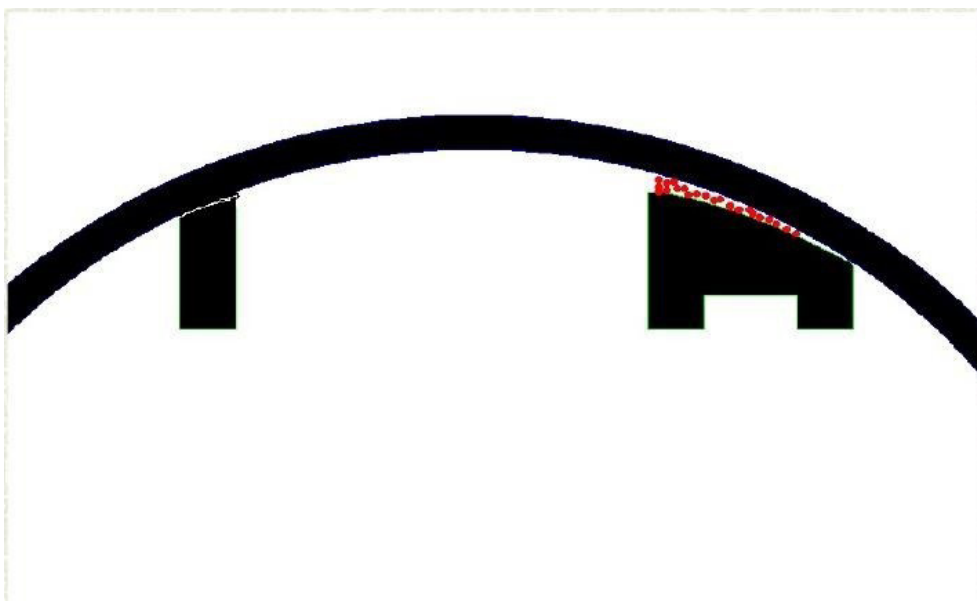


Fig 6: Showing dirt trap over a seal in suction couch roll

- Chemicals used in paper making can corrode the metal forming dirt particles.
- Sand, fillers or any other particles can enter the shell through the holes on it.
- All the above and any other particles even in dust form or through water showers can collect over the seals (as shown in red in Fig 6) causing them to wear out over the time.

FACTORS FOR SEAL MATERIAL SELECTION

Though seal life is important but the ultimate purpose of the suction couch roll seal is to fail. In case of any issue or factors listed on slide 7, if the base material of the seal does not wear out, the whole area will become hot until something else fails hence in order to prevent suction roll from damaging or any further damage to the machine it would be better if the seal wears out.

If the seal material is too hard, it might increase the life of the seal but would cause scratches on the roll whose repairing will be costlier.

Low thermal expansion of seal is vital to ensure that the seals do not exceed the gap in the holder to accommodate thermal expansion. If that happens then the seal will bend towards the shell, which will cause premature failure.

The seal material **should not absorb water** over time else it will swell in the holder and will not allow the air loading tubes to maintain proper position of the seal to hold vacuum.

In the ideal conditions the **operating temperature** of suction couch roll goes up to 80°C max. If the seal does not fail above this temperature, it will cause the damage to the rubber and PU coatings on the roll at about 100°C. Even the bond between the roll cover and shell will fail at about 120°C.

WHAT MAKES US SPECIAL?

Most wear Resistant Grade

Above 9 million molecular weight making it best wear resistant grade much higher than Graphited rubber seals.

Plascon-M has no water absorption

Plascon-M is self lubricating

In case of external lubrication failure or other reasons for strip wear (as mentioned in slide 8) Plascon- M will release its lubricants which will delay the otherwise immediate failure.

Plascon-M is a soft material

Being naturally soft, Plascon-M will wear itself without a single scratch on the roll.

Very low coefficient of Friction = Lower power consumption

Lubrication and sliding additives in Plascon-M reduce the friction which prevents local heat buildup. It also results in lower power consumption as the drag force /friction force between the shell and the strip is minimized.

Low thermal expansion.

Plascon-M seals will remain stable at ideal operating conditions. Keeping the roll safe, Plascon-M seals will deform itself on constant 95°C and above.



CONCLUSION

Understanding the importance of the role of seals in the functioning of suction couch roll, looking into why seals of a suction couch roll wear out and how Plascon- M sets apart on every factor to be kept in mind while selecting the right material for the seals, it can be very well concluded that Plascon-M is “THE” answer to all seal related problems for suction couch rolls.

